

1. Server architecture benefits. —

- Scalability - allows for easy scaling to handle increased traffic or demand.
- Reliability - A well designed server architecture ensures high uptime and minimizes downtime.
- Flexibility - It can be designed to accommodate business needs.
- Performance - Optimized server architecture improves performance.

2. Progressive Web Apps (PWAs) —

- PWAs provide a fast and responsive user experience.
- Can work offline or with a slow internet connection, allowing users to continue using the app.
- They are linkable and discoverable, makes easy for users to find.
- Can be installed on a user's device, provides native app-like experi.

3. AI and Machine Learning in software Architecture. —

- AI and ML can be used for predictive analytics, enabling software to make informed decisions.
- It can be used to personalize user experiences, tailoring the software to individual needs.
- Can automate repetitive tasks, freeing up human resources for more complex tasks.
- Can enable software to make intelligent decisions, improving overall system performance.

4. Cloud Computing Service Models —

- IaaS (Infrastructure as a Service)
 - provides virtual service, storage and network.
 - example :- Google Cloud Engine.
 - flexible and scalable.
- PaaS (Platform as a Service)
 - provides tools to develop, test and deploy.
 - example :- Google App Engine.
 - focuses only on software part and not hardware.
- SaaS (Software as a Service)
 - provides software application over internet.
 - e.g. :- Gmail, Google Docs